

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

VIth Semester of B.Sc. Biotechnology Examination May-2019

BT601 : BASIC MEDICAL BIOTECHNOLOGY

Date: 01/05/2019 Day: Wednesday Time: 01:30 pm To 01:50 pm Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
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Q - I Choose the correct answer for the following questions. 10

1. Full form of SNP is _____.
a. Single Nucleotide Polymorphism b. Stable Non-coding Polymorphism
c. Stable Nucleotide Polymorphism d. Single Non-coding Polymorphism
2. When any drug is tested on laboratory animal, then that experiment is known as _____ testing.
a. *in vitro* b. *in silico*
c. *in situ* d. *in vivo*
3. 3D structure of ligand-receptor complex can be analyzed using _____.
a. critical structure electrophoresis b. x-ray crystallography
c. infra red colorimeter d. UV-ray spectroscopy
4. Interferons (IFN) are _____.
a. enzymes b. receptors
c. cytokines d. all of the above
5. Humulin, is the brand name of _____ produced by RDT.
a. myoglobin b. tubulin
c. globulin d. insulin

6. _____ is the first chimeric and approved therapeutic antibody.
- a. Abciximab
 - b. Muromonab
 - c. Daclizumab
 - d. Murozumab
7. HAT medium allows growth of only hybrid cells due to presence of _____.
- a. HGART enzyme
 - b. HGPRT enzyme
 - c. HGMRT enzyme
 - d. HGCRT enzyme
8. DICER is an enzyme, responsible for the maturation of _____.
- a. tRNA
 - b. miRNA
 - c. hnRNA
 - d. rRNA
9. Human growth hormone consists of _____ aminoacids.
- a. 51
 - b. 24
 - c. 130
 - d. 191
10. Deficiency of Adenosine-deaminase enzyme leads to _____.
- a. severe combined immunodeficiency
 - b. cystic fibrosis
 - c. acute renal failure
 - d. neurological disorders

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BT601 : BASIC MEDICAL BIOTECHNOLOGY

Date: 01/05/2019 Day: Wednesday Time: 01:50 pm To 03:30 pm Maximum Marks: 25

Instructions:

1. **Section I and II must be attempted in TWO SEPARATE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show the necessary calculations.

SECTION - I

Q - II Answer the following questions as directed. 10

Q-IIA Answer the following questions as directed [Any Two]. 6

- 1 Mention the importance of bioinformatics and pharmacogenomics in medical biotechnology.
- 2 Write a short note on non-viral (physical, chemical) and viral methods of gene therapy.
- 3 Describe different steps of drug discovery in detail.

Q-IIIB Write a short note on any one of the following: 4

- 1 Write down different types and physiological roles of cytokines.
- 2 Stem cell therapy.

SECTION - II

Q-III Answer the following questions as directed. 15

Q-IIIA Answer any two of the following: 10

- 1 Write a short note on various fundamentals of drug design.
- 2 Explain Southern blotting technique in detail and draw a labelled diagram.
- 3 Describe different methods for the production of hormone "Insulin".

Q-IIIB Answer any one of the following: 5

- 1 Write a short note on Phase III clinical trials.
- 2 What are monoclonal antibodies? Describe their production and uses.